ENG3005 - Mechatronics/ Aerospace Design Project

Flight Control System Design for a Restricted Motion Twin Rotor UAV

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Abstract

This report presents the design, simulation, and validation of a PID-based flight control system for a restricted motion twin rotor UAV test rig. The design process follows the Systems Engineering V-model, progressing systematically from model derivation through control integration and validation. The dynamic behaviour of the rig is characterised by three rotational degrees of freedom: Elevation (ε), Pitch (θ), and Azimuth (ψ). Each of these subsystems was modelled using fundamental electromechanical relationships linking the motor input voltage to the aerodynamic thrust acting on the UAV body. The derived mathematical equations were implemented in Simulink as modular subsystems, allowing independent tuning and later combined testing under coupled motion.

The model was validated against experimental datasets provided for each axis, and parameters such as stiffness and damping were refined to align the simulated responses with the measured rig data. Following validation, three PID controllers were designed and manually tuned to maintain stable closed loop control on each axis. The elevation and pitch controllers adjusted the motor voltages directly using a mixing arrangement that combined their outputs, while the azimuth motion was controlled indirectly through its coupling with the pitch angle, reflecting the mechanical constraints of the real system. The final control setup achieved accurate tracking of reference commands during combined manoeuvre tests, showing smooth motion and keeping all actuators within safe operating limits. Overall, the results confirm that the designed control architecture provides a reliable foundation for real world implementation of the twin rotor UAV's flight control system.

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0.1 Introduction

Unmanned Aerial Vehicles (UAVs) have seen rapid growth in applications in both civilian and defense domains, driven by advances in nanotechnology, sensor technology, and autonomous control systems [1]. The effective operation of such vehicles critically depends on the design of robust flight control systems capable of regulating multi-axis dynamics in real time [2]. This report represents the design, simulation, and implementation of a PID-based flight control system for a constrained twin rotor UAV rig.

The platform utilised in this project is a twin-rotor test rig mounted on a supporting arm with a counterweight (CW). this mechanical configuration restricts the UAVs motion to three rotational Degrees of Freedom (DOF): elevation, pitch, and azimuth. Two DC motor-driven propellers generate the thrust forces required to produce motion about each axis, this setup is highly representative of the coupled, multi-input-multi-output (MIMO) dynamics encountered in practical rotor-craft design

To ensure a rigorous and traceable design process, the development of this flight control system is framed around the Systems Engineering V-model. The V-model is a highly structured framework that inherently links the initial system requirements to the subsequent verification and validation phases. In aerospace application, strictly adhering to this methodology ensures that mathematical models accurately represent the physical parameters before any code is deployed, thereby mitigating risks and reducing development costs [3].

The project follows the V-diagram design method, progressing from system specification and simulation-based controller design. Each of the three dynamic subsystems is modelled and controlled independently, then validated against raw experimental data. The combined system is constructed along with three custom PID controllers and assessed for cross-coupling effects. The physical limits of system's components are adhered to throughout the design.

0.2 Methodology

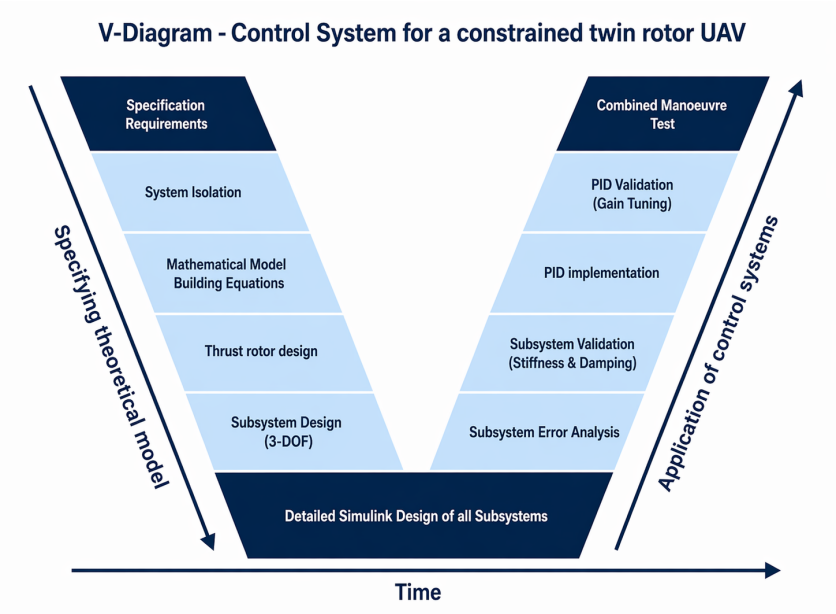


Figure 1: V-Model for System Design & Validation

The methodology for this project follows the standard Systems Engineering V-model as seen in Fig 1. This framework provides a systematic timeline for the design and validation process, mapping the progression from initial derivations down to model implementation and back up through continuous validation.

As illustrated on Fig 1, the LHS branch dictates the process designed for the construction on the Simulink models, translating the specification requirements into separate isolated subsystems. The base of the 'V' represents the core implementation phase, where any errors within the subsystems were eliminated. the RHS branch defines the Simulink test phase, validation and implementation of PID controllers. This approach follows the bottom-up trajectory, advancing from initial subsystem error analysis through to PID design, gain tuning and culminating in the final combined manoeuvre test.
Following this structured progression, the first major phase focused on establishing the underlying physics. To build the mathematical model, the specification was split into distinct subsystems to facilitate a modular approach. The electromechanical relationship was initially established between the DC motors and the Voltage input, providing with the necessary thrust force provided by the propellers. The core control of the rotors relies on the thrust equation and the 3 degrees of freedom (3-DOF) subsystems affecting the overall system stability. The 3-DOF consisting of Elevation, Pitch & Azimuth. This theoretical model represents the foundation blocks of the project, the equations derived from these physics provided a roadmap for the Simulink block diagram (BD).

0.2.1 Electromechanical Subsystem

The electromechanical subsystem models the conversion of the applied Voltage into the aerodynamic Thrust required to operate the UAV. In the Simulink BD there are two of these identical subsystems, each representing the Forward and the Back motors.

Torque Link

The fundamental bridge between the electrical input and mechanical output is the electromagnetic torque generated by the motor. This Torque (τ_M) is directly proportional to the armature current (i) and defined by the following relationship:

$$\tau_M = K_T \cdot i \tag{1}$$

Where K_T is the motor torque constant. For the chosen Pittman 9X34 DC motor (12V winding), the constant specified for this project is $K_T = 0.0182 N m A^{-1}$.

Electrical equation

The electrical behaviour of the motor circuit is governed by Kirchhoff's Voltage Law (KVL). The applied motor Voltage, V(t), must overcome the armature resistence (R), the armature inductance (L) and the back electromotive force (EMF). Applying this to the specified physics have yielded the following expression:

$$\frac{di}{dt} = \frac{1}{L_a}(V - R_m \cdot i - K_e \cdot \omega_m) \tag{2}$$

- V is the received voltage measured in V.
- $R_m = 0.83\Omega$ armature resistance.
- i is the current measured in A .
- $L_a = 0.63mH$ is the armature inductance measured in mH
- $\frac{di}{dt}$ the rate of change in current
- $K_e = 0.0182\frac{V}{rad/s}$ The back EMF.
- ω is the angular velocity of the motor shaft (rad/s)

As the motor accelerates, the back EMF term ($K_e \cdot \omega$) increases, which leads to a decrease in the current flowing through the circuit. This provides the self–regulating mechanism meaning the motor limits its own speed.

Mechanical equation

Motor shaft represented by applying Newton's 2nd law for rotation:

$$(\frac{d\omega}{dt}) = \frac{1}{J_m}(K_t \cdot i - B_{sm} \cdot \omega_m) \tag{3}$$

where:

- $J_m = 1.91e^{-6}kg.m^2$ is the motor rotor combined moment of inertia.
- $\frac{d\omega}{dt}$ this is the angular acceleration of the shaft unit:$\frac{V}{rad/s}$.
- $K_t = 0.0182Nm/A$ is the torque constant .
- $B_{sm} = 0.0002Nm/\frac{rad}{s}$ this is the viscus damping coefficient of the motor shaft.

The torque constant K_t and the K_e back EMF constant have the same value usually because DC motors are known for converting between electrical and mechanical domains.

Both of these equations are combined: Electrical equation produces current that feeds into the Mechanical expression to generate torque and angular velocity ω, which feeds back to the Electrical equation as back–EMF. This coupling is illustrated in both motors for both the Forward and Backward Simulink block diagrams as in Figure 2.

Aerodynamic Thrust Conversion

The final stage of the electromechanical subsystem converts the angular velocity (ω) of the motor shaft into aerodynamic thrust force produced by the propeller. Based on standard propeller theory, this relationship is defined by the full aerodynamic thrust equation:

$$F = C_t \cdot \rho \cdot n^2 \cdot D^4 \tag{4}$$

where:

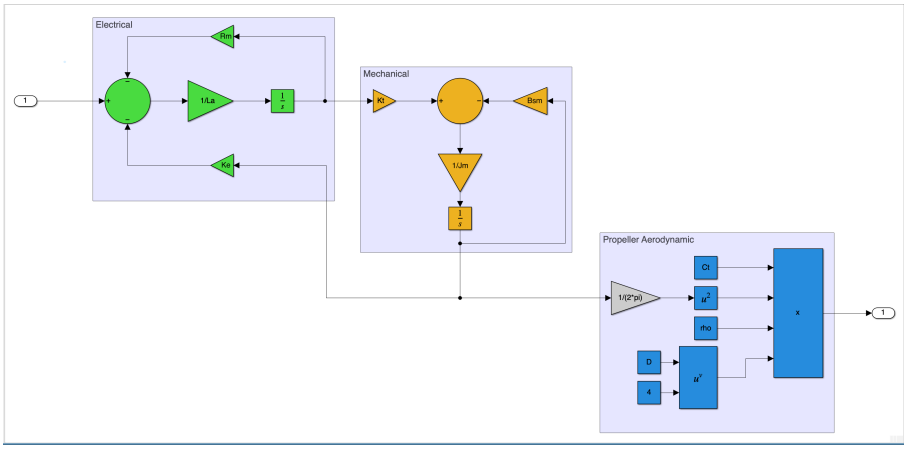


Figure 2: Thrust Subsystem BD, showing the electrical motor & mechanical rotor dynamics, and propeller aerodynamic thrust model.

- F is the thrust force produced by the propeller (N)
- $C_t = 0.065$ is the thrust coefficient
- $\rho = 1.225 kg/m^3$
- n (rev/s) propeller rotational speed
- $D = 0.2m$ propeller diameter

To maintain adherence to SI units for the mechanical dynamics, the motor's angular velocity (ω) in rad/s is converted within the subsystem using $n = \omega/2\pi$. Substituting this into the Thrust equation yields the final aerodynamic output. This complete sequence, from electrical voltage to physical Thrust, is illustrated in the subsystem block diagram shown in Fig 2.

0.2.2 3-DOF Subsystems

The rotational motion of the UAV is decomposed into three separate DOF:**Elevation(ϵ), Azimuth(ψ)** and **Pitch(θ)**. The dynamics of each axis are governed by Newton's 2nd) Law for rotational motion.

$$\sum \tau = I \cdot \alpha \tag{5}$$

The sum of Moments equals to the product of the moment of inertia (I) and angular accelaration (α). For each subsystem, the applied aerodynamic thrust moments must overcome the opposing moments from gravity, structural stiffness and viscous damping, in order to capture the true mechanical behaviour of the rig.

Elevation

The elevation angle ϵ governs the altitude motion of the UAV, rotating about the central horizontal hinge. The motion is driven by the combined Thrust of the forward and back motor ($F_F + F_B$), with gravity acting in the opposite direction to the elevation force.

Newton's 2nd law for rotation about the Y axis:

$$\sum M_{Elevation} = J_E\ddot{\epsilon} = M_{Thrust,\epsilon} - M_{Gravity} - M_{Damping} - M_{Stiffness} \tag{6}$$

$$M_{Thrust,\epsilon} = l_A \cdot K_F(F_F + F_B) \cdot cos\theta \tag{7}$$

Expand and Rearrange:

$$\ddot{\epsilon} = \frac{K_F}{J_E} \cdot l_A(F_F + F_B) \cdot cos\theta - \frac{1}{J_E}(m_{UAV} \cdot g \cdot l_A - m_{CW} \cdot g \cdot l_{CW}) \cdot cos\epsilon - \frac{B_E}{J_E}\dot{\epsilon} - \frac{K_E}{J_E} \cdot \epsilon \tag{8}$$

J_E Moment of inertia about the elevation axis, 0.91 kg·m^2

$\ddot{\epsilon}$ Angular acceleration about the elevation axis (rad/s^2)

$\dot{\epsilon}$ Angular velocity about the elevation axis (rad/s)

ϵ Elevation angle (rad)

l_A Arm length, 0.660 m

F_F, F_B Forward and backward thrust forces (N)

cos(θ) Reduction in vertical thrust due to pitch

$m_{\mathbf{UAV}}$ Mass of the UAV, 1.525 kg

$m_{\mathbf{CW}}$ Mass of the counterweight, 1.87 kg

$l_{\mathbf{CW}}$ Arm length to the counterweight, 0.470 m

g Gravitational acceleration, 9.81 m/s^2

B_E Elevation damping coefficient, 0.58 Nm/(rad/s)

K_E Elevation stiffness, 2.548 Nm/rad

Azimuth

The Azimuth angle ψ describes the horizontal rotational motion of the UAV arm around the central vertical base axis. It is the heavily coupled force out of the 3-DOF. The torque motion is driven from the total thrust ($F_F + F_B$) and the moment is only executed when the Pitch(θ) angle is non–Zero. This happens because the thrust acts along the propeller axis, and the Pitch motion tilts to redirect the thrust movement of the arm into the horizontal plane. The Elevation angle (ϵ) affects the horizontal moment arm via a cosine relationship.

Newton's 2nd law for rotation about the Z axis:

$$\sum M_{\text{Azimuth}} = J_A\ddot{\psi} = M_{\text{Thrust},\psi} - M_{\text{Damping}} \tag{9}$$

$$M_{Thrust,\psi} = l_A(F_F + F_B) \cdot sin\theta \cdot cos\epsilon \tag{10}$$

Expand and Rearrange:

$$\ddot{\psi} = \frac{1}{J_A} \cdot l_A(F_F + F_B) \cdot sin\theta \cdot cos\epsilon - \frac{B_A}{J_A}\dot{\psi} - \frac{K_A}{J_A} \cdot \psi \tag{11}$$

J_A Moment of inertia about the azimuth axis, 0.91 kg·m^2

l_A Arm length from the elevation axis to the UAV body, 0.660 m

F_F, F_B Forward and backward thrust forces (N)

θ Pitch angle (rad)

ϵ Elevation angle (rad)

$\ddot{\psi}$ Angular acceleration about the azimuth axis (rad/s^2)

$\dot{\psi}$ Angular velocity about the azimuth axis (rad/s)

ψ Azimuth angle (rad)

B_A Azimuth damping coefficient, 0.016 Nm/(rad/s)

K_A Azimuth stiffness, 0.1e^-6 Nm/rad

Pitch

The Pitch angle θ represent the tilt of the UAV rotor assembly about the hinges connecting it to the main supporting arm. Unlike Elevation and Azimuth, which are driven by the sum of the Thrust Forces, the Pitch motion is exclusively driven by the differential Thrust between the Back and Forward motors.

Newton's 2nd law for rotation about the X axis:

$$\sum M_{Pitch} = J_P \cdot \ddot{\theta} = M_{Thrust,\theta} - M_{Damping} - M_{Stiffness} \tag{12}$$

$$M_{Thrust,\theta} = l_A \cdot K_F(F_B - F_F) \tag{13}$$

Expand and Rearrange:

$$\ddot{\theta} = \frac{K_F}{J_P} \cdot l_A(F_B - F_F) - \frac{B_P}{J_P}\dot{\theta} - \frac{K_P}{J_P}\theta \tag{14}$$

where:

J_p Moment of inertia about the pitch axis, 0.0364 kg·m^2

K_F Thrust constant, 0.1188 N/V

l_A Distance from the pitch axis to either motor, 0.178 m

F_B, F_F Backward and forward thrust forces (N)

$\ddot{\theta}$ Angular acceleration about the pitch axis (rad/s^2)

$\dot{\theta}$ Angular velocity about the pitch axis (rad/s)

θ Pitch angle (rad)

B_P Pitch damping coefficient, 0.00659 Nm/(rad/s)

K_P Pitch stiffness, 0.287 Nm/rad

0.3 Simulation Model

The derived 3-DOF dynamic equations were implemented in Simulink using subsystem block diagrams for each rotational axis. Figures 3, 4, and 5 show the implementation of the elevation, azimuth, and pitch subsystems respectively. Each model follows the structure of summing moments, dividing by inertia, and applying double integration to obtain angular motion.

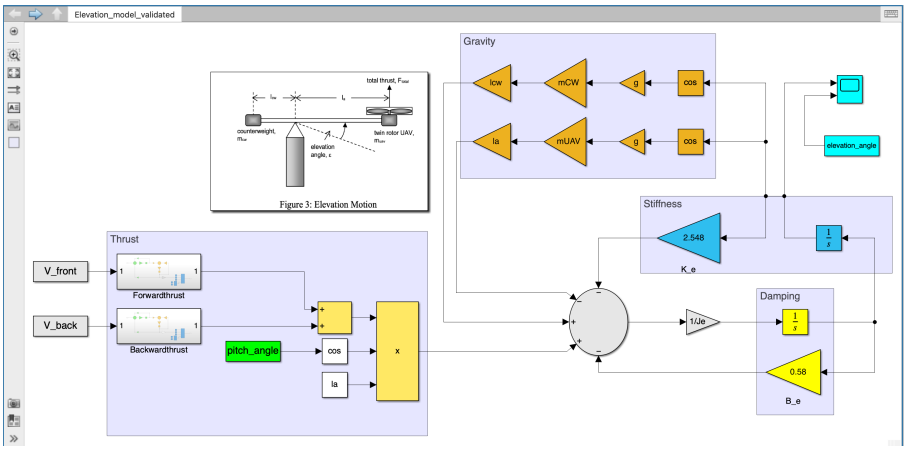


Figure 3: Elevation BD showing thrust summation, gravitational effects, & damping/stiffness feedback.

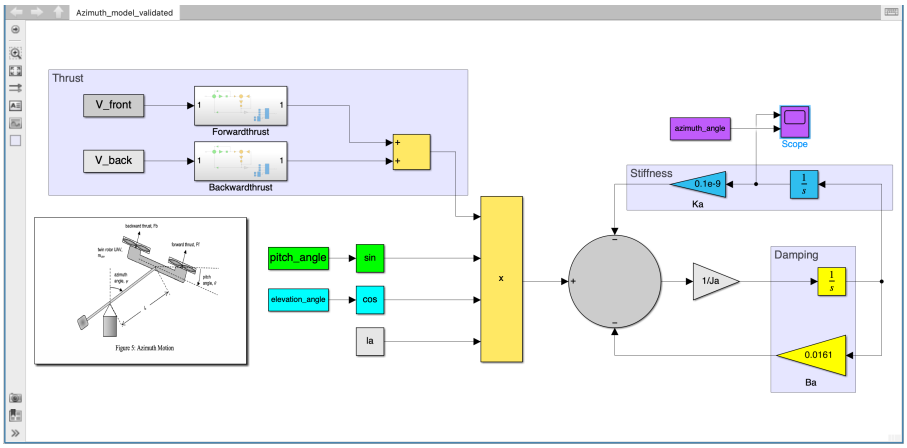


Figure 4: Azimuth BD illustrating non-linear coupling through $\sin\theta$ & $\cos\epsilon$ terms.

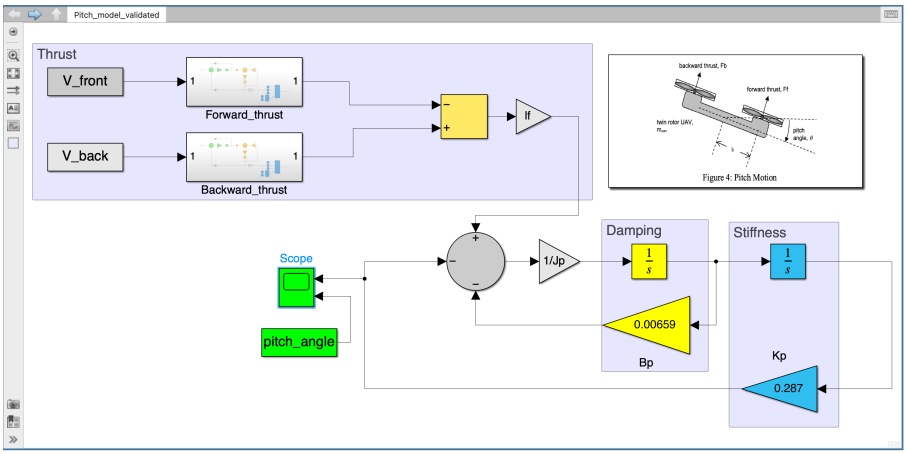


Figure 5: Pitch BD driven by differential thrust & including damping and stiffness effects.

0.3.1 Physical Model

Once the equations for elevation, pitch, and azimuth were derived, each subsystem was built in Simulink by rearranging the equations into explicit angular acceleration form. This made the model fairly straightforward to put together using standard dynamic blocks, since Simulink can treat each rotational degree of freedom as a second-order system—angular acceleration comes out as an intermediate output, and then angular velocity and displacement follow from successive integration. So the overall structure of each subsystem was based directly on the expressions for $\ddot{\epsilon}$, $\ddot{\theta}$, and $\ddot{\psi}$.

The first step was converting the rotor input voltages into thrust signals for the front and back propellers. These thrust signals were then either summed or subtracted depending on what was needed. The appropriate geometric coupling terms—like $\cos\theta$, $\cos\epsilon$, or $\sin\theta$—were included depending on which physical motion was being modelled. Damping and stiffness effects came in as negative feedback terms, proportional to angular velocity and angular displacement respectively. The total moment acting about each axis was then divided by the relevant moment of inertia to get angular acceleration, and two cascaded integrator blocks gave the angular velocity and position.

Thrust Block Implementation

The first stage of each subsystem was the thrust generation block. The front and back motor voltages, V_{front} and V_{back}, were passed through gain blocks representing the thrust conversion, which produced the individual thrust signals F_F and F_B. In the Simulink model, these showed up as separate forward-thrust and backward-thrust paths before feeding into the 3-DOF dynamic subsystems. How these thrust signals were used depended on the axis being controlled. For elevation and azimuth, the total thrust $(F_F + F_B)$ was needed, while for pitch, the driving input was the differential thrust $(F_B - F_F)$. This made the thrust block the common interface between the electromechanical model and the rigid-body rotational dynamics.

Elevation Subsystem Implementation

The elevation subsystem was built from the derived rotational equation by forming the total thrust term and converting it into a moment about the elevation axis. The summed thrust $(F_F + F_B)$ was multiplied by the arm length l_a and by the pitch coupling term $\cos\theta$, giving the thrust contribution to elevation motion. A second branch represented the gravitational moment from the imbalance

between the UAV mass and the counterweight, scaled by $\cos\epsilon$. Damping and stiffness were included as feedback terms using the elevation rate and angle signals, multiplied by the constants B_E and K_E respectively. These contributions were summed algebraically and then multiplied by $\frac{1}{J_E}$ to give the angular acceleration $\ddot{\epsilon}$. Two integrator blocks in series then produced $\dot{\epsilon}$ and ϵ.

In block-diagram form, the subsystem followed the structure:

$$(F_F + F_B) \rightarrow \text{moment calculation} \rightarrow \sum M_E \rightarrow \frac{1}{J_E} \rightarrow \ddot{\epsilon} \rightarrow \int \rightarrow \dot{\epsilon} \rightarrow \int \rightarrow \epsilon$$

This arrangement kept the physical contributors to elevation motion clearly identifiable within the Simulink model.

Azimuth Subsystem Implementation

The azimuth subsystem needed the most coupling and was therefore the most nonlinear of the three. The total thrust term $(F_F + F_B)$ was formed first, since azimuth motion depends on the combined rotor force. This signal was then multiplied by $\sin\theta$ and $\cos\epsilon$ using trigonometric function blocks and product blocks. Physically, these terms represent the redirection of rotor thrust into the horizontal plane when the vehicle is pitched, while also accounting for how the elevation angle affects the available horizontal component. The azimuth damping and stiffness terms were implemented in the same way as the other subsystems, using feedback from $\dot{\psi}$ and ψ through gain blocks B_A and K_A. The total azimuth moment was then divided by the azimuth inertia J_A to get $\ddot{\psi}$, which was integrated twice to produce azimuth rate and angle.

This implementation preserved the strong coupling behaviour of the physical system. In particular, when $\theta = 0$, the $\sin\theta$ term becomes zero and no azimuth motion is generated—exactly matching the expected behaviour of the real rig.

Pitch Subsystem Implementation

The pitch subsystem was simpler to build because it didn't include a gravitational moment term. The front and back thrust signals were passed into a subtraction block to form the differential thrust $(F_B - F_F)$, which is the driving input for pitch motion. This signal was then multiplied by the pitch arm length l_f to generate the pitch moment. Damping and stiffness were implemented as negative feedback loops using gain blocks on $\dot{\theta}$ and θ. The resulting net moment was multiplied by $\frac{1}{J_P}$ to get the pitch angular acceleration $\ddot{\theta}$. As with elevation, two cascaded integrators then gave pitch rate and pitch angle.

This gave the pitch subsystem a compact and physically clear structure—the thrust difference drives the motion, and the damping and stiffness terms oppose it through feedback. The implementation directly reflected the derived second-order equation of motion.

Overall Modelling Approach

Although the elevation, pitch, and azimuth subsystems were developed as separate Simulink blocks, they weren't independent in operation. Signals like pitch angle and elevation angle were routed between subsystems to provide the geometric coupling terms required by the nonlinear model. As a result,

the final implementation kept the physical relationships identified during the derivation stage, while still allowing each axis to be developed and tested in a modular way. This approach made the model both analytically traceable and practical to implement in Simulink as part of the full twin-rotor UAV control system.

0.4 Validation

0.4.1 Subsystem Validation

Firstly, the functionality of each subsystem was confirmed by validating each 3-DOF axis against experimental data provided for validation purposes. The validation data **.mat** files contained the recorded motor voltages and the corresponding angle measurements from the rig's sensors. After ensuring the subsystems functioned without errors after being fed constant values, the validation signals were fed into each Simulink model as inputs by adding ***From Workspace*** blocks. Thus, the simulated output signals was overlaid the Validation graphs inside of the scope sink.

Empirical Response Analysis

The responses of the validation signal were empirically analysed to extract benchmark values for the stiffness and dampness coefficients of each axis. By observing the tail end of each response, when all transients had decayed, thus reaching the steady-state. These steady-state values were identified and used to verify the gravitational torque balance.
For the elevation axis, the ring-down region of the step response was isolated and the DC offset removed to centre the oscillation abo zero. Successive peaks were identified to be used for computation of the logarithmic decrement.

$$\delta = \frac{1}{N} \ln \frac{x_1}{x_{N+1}} \tag{15}$$

Where x_1 & x_{N+1} correspond to respective peak amplitudes over the N-number of cycles. The δ value was required in order to find the systems damping ratio (ζ).

$$\zeta = \frac{\delta}{\sqrt{1-\zeta^2}} \tag{16}$$

Thanks to this and using the mean period of successive peaks (T_d), the remaining properties of the validation response could be found:

$$\omega_d = \frac{2\pi}{T_d}, \qquad \omega_n = \frac{\omega_d}{\sqrt{1-\zeta^2}} \tag{17}$$

With these system parameters worked out and the moment of inertia being known ($J_e = 0.91 kgm^2$), the stiffness and damping benchmarks for elevation can be worked out by using the following expressions:

$$K_e = J_e \cdot \omega_n^2, \qquad B_e = 2\zeta\omega_n \cdot J_e \tag{18}$$

For elevation, these values were found to be $K_e = 2.548Nm \cdot rad^{-1}$ and $B_e = 0.58Nm/rad \cdot s^2$.
The azimuth axis exhibits no oscillatory behaviour, instead the response is overdamped and increases with the applied Voltage. Therefore its damping coefficient was estimated from the steady state Torque balance. The driving Torque during the ramp was computed as $(tau = l_a(F_f + F_b)\sin\theta\cos\epsilon$, and since this is in steady state, it must equal the damping Torque:

$$B_a = \frac{\bar{\tau}}{\dot{\bar{\psi}}} \tag{19}$$

The over bar present in the above expression denotes the averages taken over the steadystate window $(45 - 60s)$. This gave the benchmark value for damping, $B_a = 0.06068NM/rad \cdot s^{-1}$.
for pitch, the irregular nature of the response given by the validation data, made the logarithmic decrement method an unreliable tool for the analysis. The oscillations were of varying amplitude and frequency, so instead of calculating benchmark values, the values for the damping and stiffness of the system were determined by trial and error.

0.4.2 Validation Results

Elevation

The elevation validation was performed using the *Elevation_step_4V_set_1.mat* dataset, where both motors received a constant $4V$ step input. Although the benchmark values were calculated to be $K_e = 2.548Nm/rad$ and $B_e = 0.58NM/rad \cdot s^{-1}$, these only provided a good starting point. As shown in Fig 6 the benchmark response captures the general oscillation frequency very well, however, the system settles lower than the plot given by the validation data which settles at $-0.4rad$.

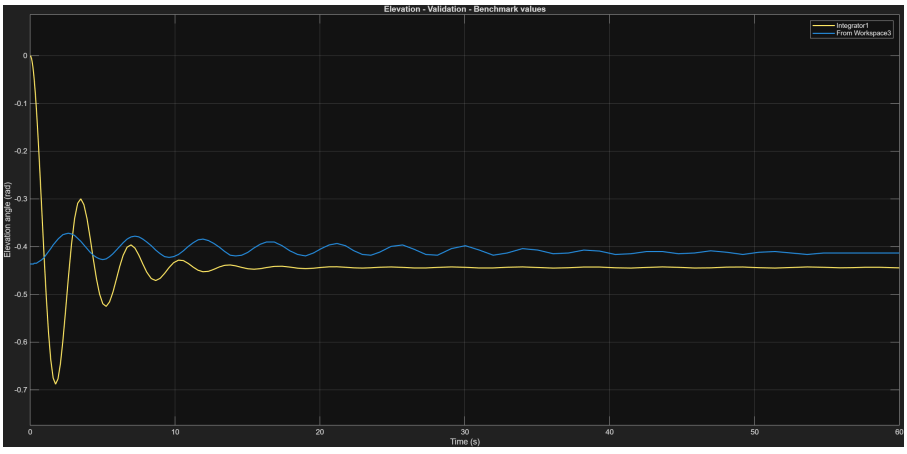


Figure 6: Elevation system response using benchmark values $K_e = 2.548Nm/rad$ and $B_e = 0.58NM/rad \cdot s^{-1}$

to correct this, the stiffness and damping values were manually tweaked iteratively, they were analysed in isolation to determine whether reducing the value increased or decreased the final steady-state value, and adjusted the damping to ensure that the oscillation stayed within the acceptable range. The final evelated values were determined to be $K_e = 2.77NM/rad$ and $B_e = 0.39Nm/rad \cdot s^{-1}$. As illustrated in Fig 7, the tuned model shows a much closer match to the experimental response throughout the

full $60s$. The oscillation decay rate, the damped frequency and the steady-state value of $-0.4rad$ all align well between the two signals.

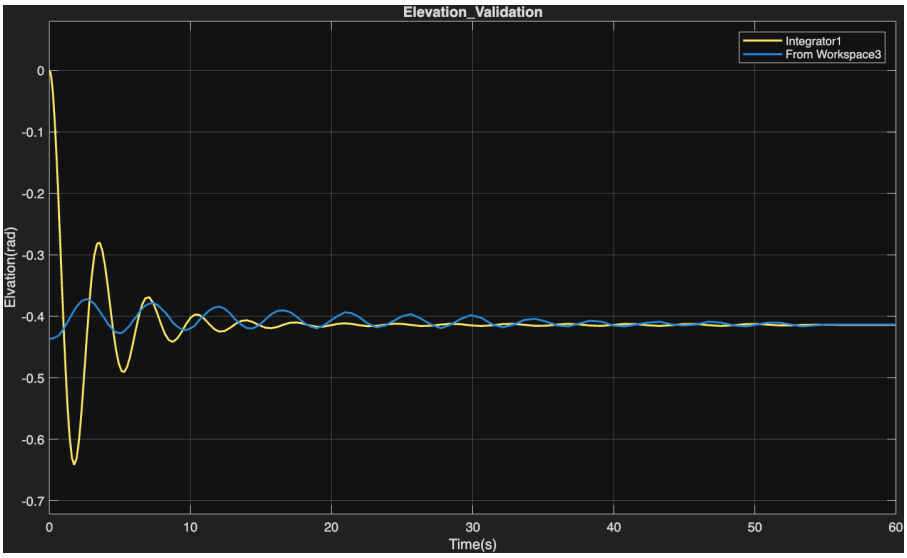


Figure 7: Elevation Validation values $K_e = 2.548Nm/rad$ & $B_e = 0.58NM/rad \cdot s^{-1}$.

Azimuth

The Azimuth validation preformed using the (*Azimuth_stepwave_7to8V_set_1.mat*) dataset.The steady-state Torque balance method yielded a decent benchmark, as shown in Fig 8. this value required further tuning because it still fell short of reaching approximately the $25rad$ mark by $60s$. This suggested that the benchmark damping value was too high and needed to be reduced in further tuning.

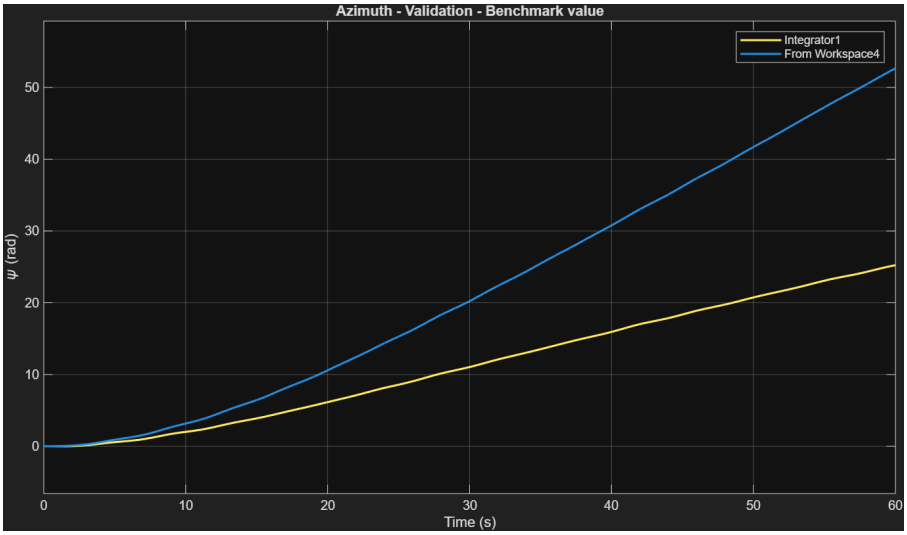


Figure 8: Azimuth system response using the benchmark value $B_a = 0.06068$

Through iterative reudction of the damping coefficient, the final validated value was $B_a = 0.0161Nm/rad \cdot s^{-1}$. As shown in Fig 9 this final tuned simulation tracks the experimental ramp nearly perfectly, with both curves reaching the approximate $52rad$ by $60s$. The small offset between the two signals persists,

likely attributable to non-linearities which weren't taken into the account in the Simulink model. The near-zero stiffness reflects the physical reality of the azimuth axis, unlike elevation, there is no gravitational restoring force acting to return the arm to its natural position after being displaced. The Azimuth axis is free to rotate in either direction, meaning that any Torque which is applied accelerated the arm instead of being opposed by a spring-like resistance. Setting the $K_a = 0.000001 Nm/rad$, which is a negligible value, avoids potential numerical issues in the model whilst still faithfully representing this behaviour.

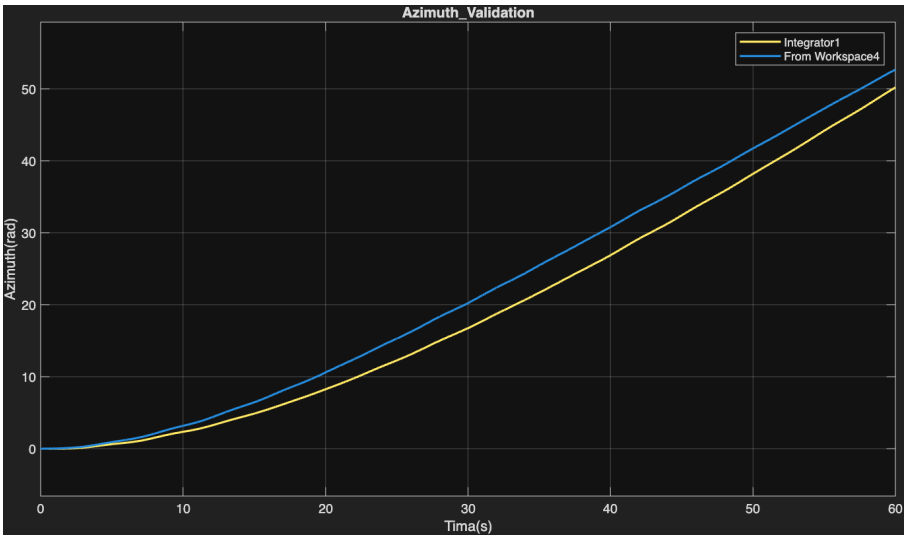


Figure 9: Azimuth Validation value $B_a = 0.0161 Nm/rad \cdot s^{-1}$

Pitch - Case 1

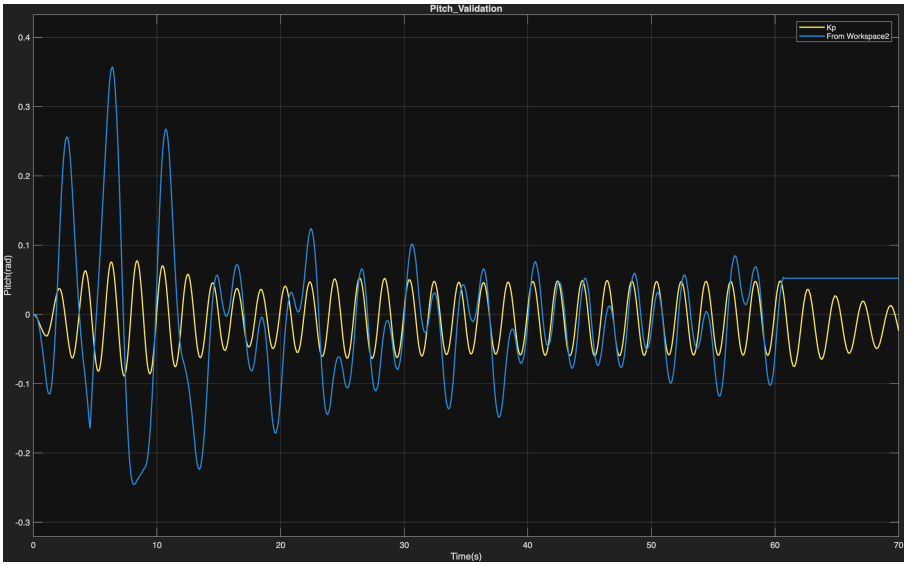


Figure 10: Pitch Validation value $K_p = 0.287 Nm/rad$ & $B_p = 0.0065 NM/rad \cdot s^{-1}$.

For Pitch validation preformed using the (*Pitch_stepwave_7to8V_set_1.mat*) dataset, where front motor received $7V - to - 8V$ square wave and for the back motor the inverse $8V - to - 7V$. As previously notes in the empirical response analysis, the irregular nature of the pitch response made benchmark

estimation unreliable, hence the coefficients were determined through manual trial and error, there are two cases for the Simulation response both presented below.
Case 1 is illustrated in Figure 10, where $K_p = 0.287Nm/rad$ & $B_p = 0.0065NM/rad \cdot s^{-1}$, in this example the simulated response matches the oscillation frequency very well throughout the full duration, however, the very low damping coefficient fails to settle as a steady straight line near the value of $0.052rad$. This indicated that the model is significantly underdamped relative to the real system.

Pitch - Case 2

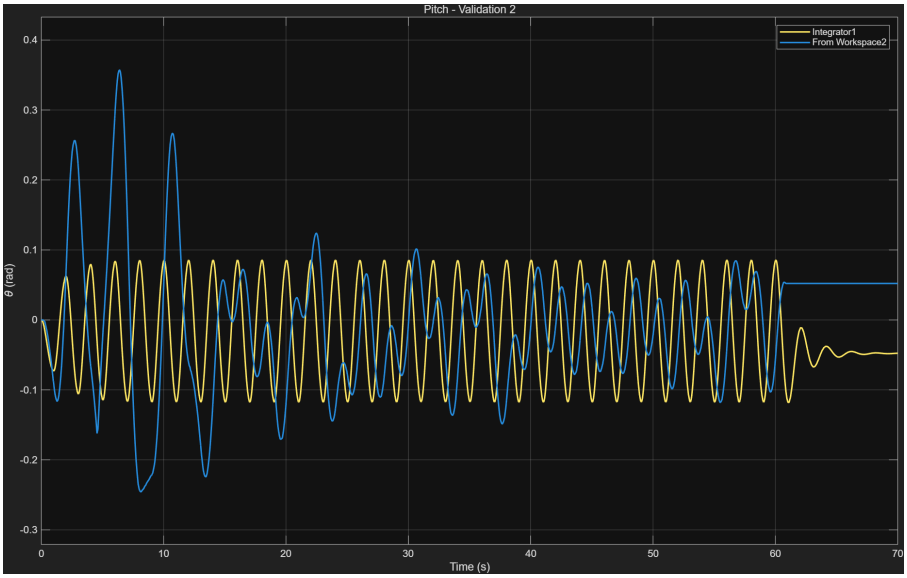


Figure 11: Pitch Validation $K_p = 0.35Nm/rad$ & $B_p = 0.045NM/rad \cdot s^{-1}$

Following Case 1, in this version the damping coefficient was increased substantially with the stiffness needing only a slight adjustment. With $K_p = 0.35Nm/rad$ & $B_p = 0.045NM/rad \cdot s^{-1}$ the simulated result managed to capture a decent balance between both, the scillation frequency and the amplitude, along with the decay behaviour. As seen on Fig 11 both of the signals settle at approximately $0.05rad$ by around $65s$.

0.4.3 PID Controller Design

The Final three PID controllers were designed to regulate the Elevation, Azimuth and Pitch axes.
The standard PID control Formula:

$$u(t) = K_p \cdot e(t) + K_i \cdot \int (e(t)) + K_d \cdot \frac{de(t)}{dt} \tag{20}$$

$e(t)$ The `error` between the measured output and the reference command.

K_P The proportional gain.

K_i The integral gain.

K_d The derivative gain.

The voltage mixing PID output connects to the motor voltage as follows:

$$V_F = PID_{Elevation} + PID_{Pitch} \tag{21}$$

$$V_B = PID_{Elevation} - PID_{Pitch} \tag{22}$$

Figure 12 represent the full PID's connection the PID_Elevation, PID_Azimuth and PID_Pitch. **PID_Elevation output:**controls the average voltage applied to both motors, which determines the thrust total and drives the elevation motion. **PID_Pitch output**:controls the difference in voltage between the motors,that creates a differential thrust and tilts the UAV in Pitch. **PID_Azimuth output**: Azimuth's output feeds into pitch reference input, it dose not connect directly to the voltages. When Azimuth error exists its controller commands Pitch angle to produces a $sin(\theta)$ term in Azimuth thrust equation (9)**, which generate the horizontal torque needed to rotate the arm. This connection reflects the physical application of the system, where the only possible way to control the Azimuth motion is via Pitch coupling.
The reference commands provided through the *(elev_ref) - and - (azi_ref)* workspace variables for both Elevation and Azimuth as shown in Figure 12. Pitch driven entirely by the azimuth controller's demand, thus why its has no independent reference.

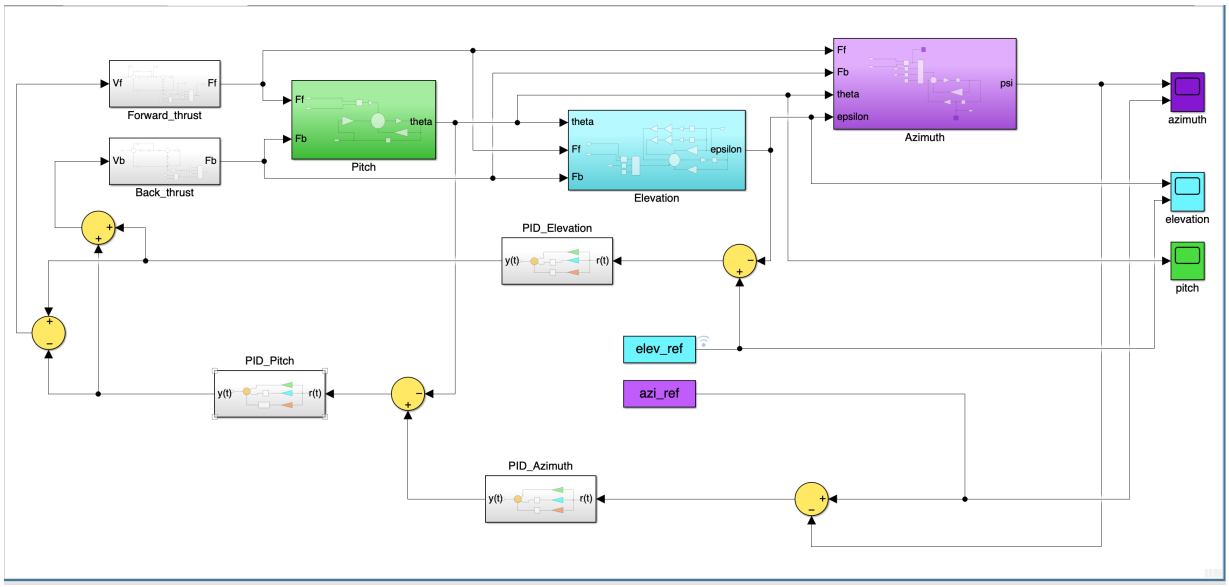


Figure 12: Full structured PID block diagram

0.4.4 Tested PID configuration:

In the pursuit of obtaining the most accurate values that match the reference wave, a second experimental approach has been followed, including the standard one, which is using the Derivative block diagram in all three PIDs; this does not execute the desired values no matter how much alteration has been applied on the PID gains values. Meanwhile, using the Transfer Function Block diagram in PID_Pitch does.

Case1: Using Transfer Function Block on Pitch only

In this module, the pitch PID derivative term uses a filtered derivative block with transfer function $\{\frac{1}{(s+1)}\}$ instead of a pure derivative. This adds a low pass filter to the derivative action, which reduces sensitivity to high frequency noise in the pitch signal. Where the elevation and azimuth controllers uses pure derivative blocks as shown in figure 13.

The gain values for Case 1 are:

Controller	Kp	Ki	Kd
Elevation	5.5	5.5	1.0
Pitch	0.007	0.006	0.690
Azimuth	0.45	0.0215	6.65

Table 1: Gain values for Case 1

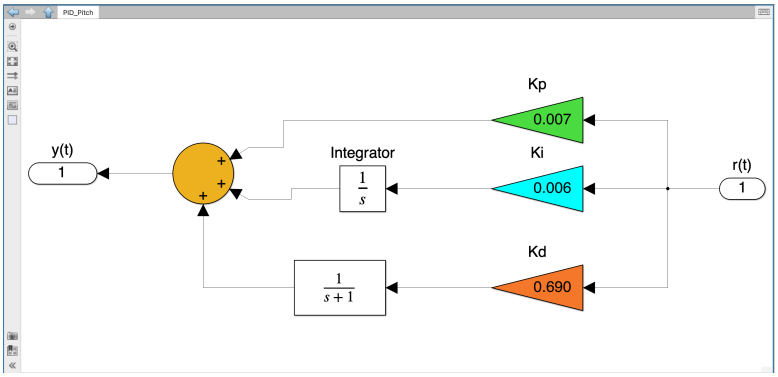


Figure 13: Transfer Function used instead of derivative at PID_Pitch block diagram

Case2: Pure Derivative Blocks throughout all PIDs

Here all three PID controllers use pure derivative blocks $\frac{du}{dt}$ for the derivative term. No filtering is applied, Figure 14.

The gain values for Case 2 are:

Controller	Kp	Ki	Kd
Elevation	2.5	8	4.5
Pitch	3.5	0.015	0.0015
Azimuth	0.35	0.0025	0.004

Table 2: Gain values for Case 2

The gains in both cases determined by combination of manual tuning and blocks placement analysis. The initial values founded by manually tuning the gains while observing the simulation output response. These values were than compared against MATLAB script calculations, which provided closes values to the manually toned ones. However, the manually tuned values produced overall better tracking performance and were selected as the final gains values.

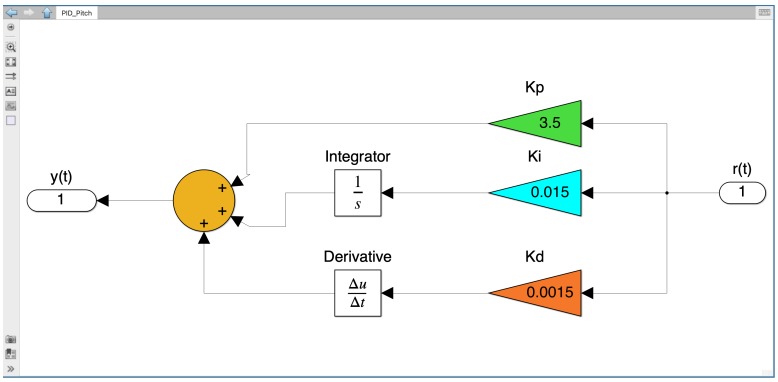


Figure 14: Pure derivative block diagram has been used through out the PIDs

0.4.5 PID Validation

Both PID_Elevation and PID_Azimuth were validated first using step responses, then using the combined manoeuvre that commands both simultaneously with step–time reference of 330 seconds.

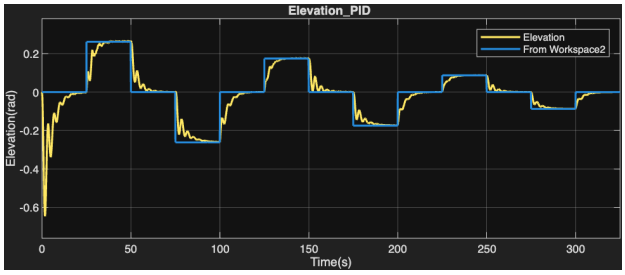
Case 1: PID Using Transfer function for Pitch only

In **Case 1**, elevation controller tracks the reference commands accurately with small overshoot and quick settling Figure 15a. Pitch signal shows the expected oscillatory response as the Azimuth controller commands Pitch angles to lead azimuth motion Figure 15c. While the Azimuth response tracks the reference very well, by following each step with acceptable settling time, Figure 15b and
Case1: Combined Manoeuvre test shows quite perfect tracking performance on both the elevation and azimuth axes as in Figure 15d.

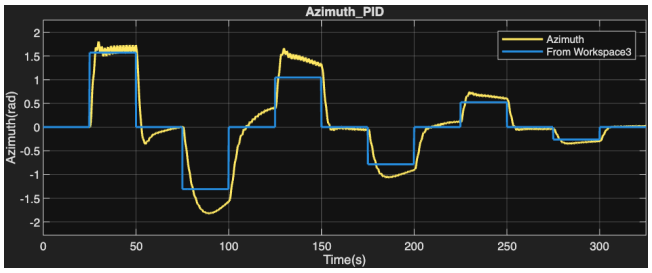
Case 2: PID using Pure derivative only

In **Case 2**, elevation controller results similar to Case 1 for individual step tests, look at Figure 16a. Pitch response was stable due to the different gain values. The Azimuth individual tests show slower tracking compared to Case 1; it diverges significantly from the reference with output overshoot and failing to return to the commanded positions as in Figure 16b. This divergence is caused by the derivative block amplifying noise spicks in the pitch signal Figure 16c , which creates an unstable feedback loop that derives the azimuth controller during complex manoeuvre.
Case2: Combined Manoeuvre test shows that elevation tracking remains acceptable as shown in Figure 16d.

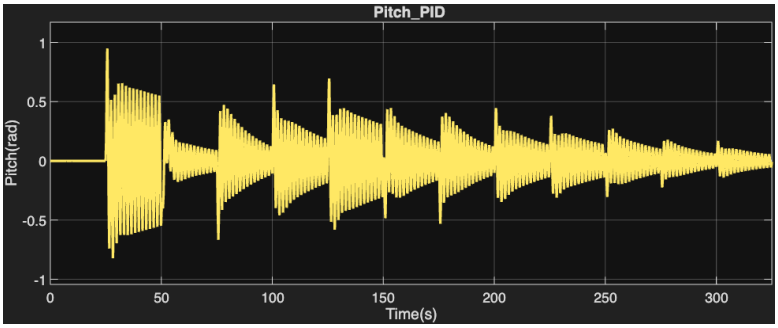
Case1: was selected as the final controller design because it provided stable, and accurate tracking across all three axes and in the combined manoeuvre test. The filtered derivative on the pitch PID was the key difference that improved robustness.



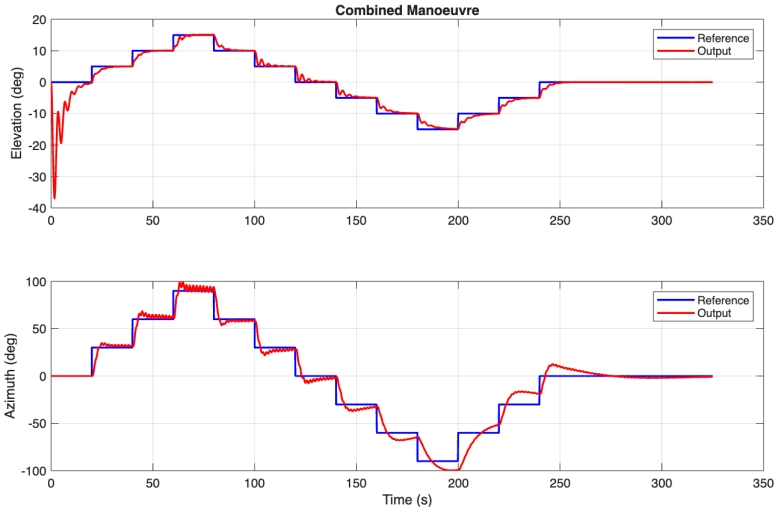
(a) Case1- PID_Elevation system response



(b) Case1- PID_Azimuth system response

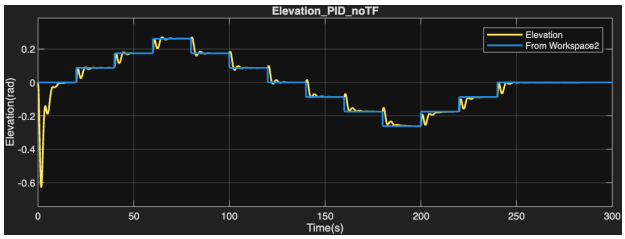


(c) Case1- PID_Pitch system response

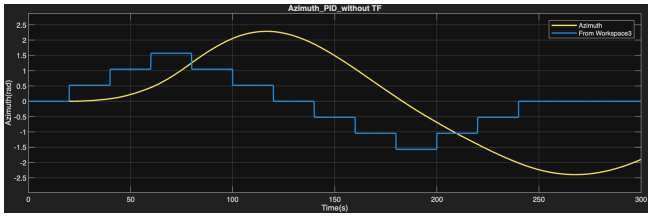


(d) Case1- Combined Manoeuvre

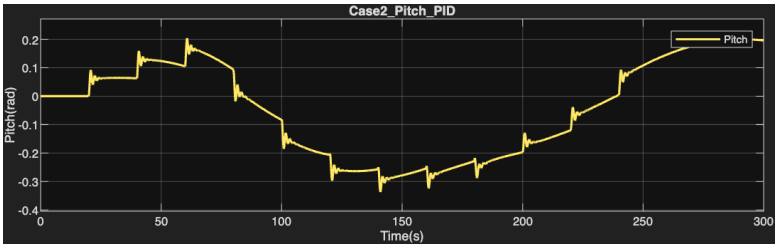
Figure 15: Case1- PID Three axes output using Transfer function block only for Pitch PID



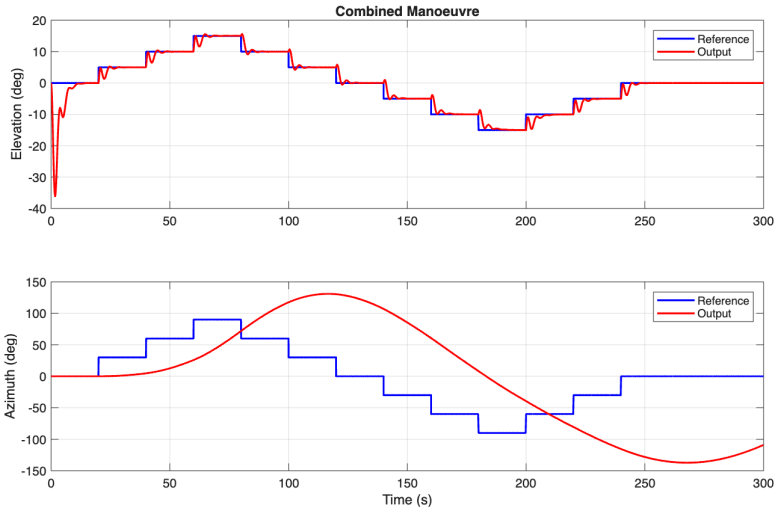
(a) Case2- PID_Elevation system response



(b) Case2- PID_Azimuth system response



(c) Case2- PID_Pitch system response



(d) Case2- Combined Manoeuvre

Figure 16: Case2- PID Three axes output using Derivative blocks only

0.5 Results & Discussion

0.5.1 Model Analysis

The simulation model was evaluated cross all three axes by comparing open loop Simulink responses against the experimental reference validation datasets. All subsystems been tested individually before integrating it into the full wired model.

The **Elevation subsystem**, the benchmark parameter execution using logarithmic decrement analysis yielded initial values of $K_e = 2.548$ Nm/rad and $Be = 0.58 Nm/rad.s^{-1}$. These values captured the oscillation frequency accurately, but the steady state simulation settled lower than the measured $-0.4 rad$ from the physical rig. Manual tuning produced refined values of $K_e = 2.77$ Nm/rad and $Be = 0.39$ Nm/rad.s^{-1}, with these values the model aligned well with the experimental data in terms of decay rate, damped frequency and steady state position.

The **Azimuth subsystem** presents different challenge, the axis shows no oscillatory behaviour and responds as an over–damped ramp constant voltage input. The benchmark value of $Ba = 0.06068 Nm/rad^{-1}$, this derived from the steady state torque balance, that caused the simulation to fall short of the $53 rad$ reached by the physical rig at $60 s$. This illustrate the benchmark was significantly overestimating the true rotational resistance. The final validated value of $Ba = 0.0161 Nm/rad.s^{-1}$ generate near prefect ramp tracking. Stiffness value of $Ka = 0.000001 Nm/rad$ reflects the physical absence of any gravitational restoring force on the azimuth axis.

The **Pitch subsystem** and axis was the most difficult to validate analytically. The irregular oscillations and frequencies of the experimental pitch response leads to unreliable tuning method, so so parameters were chosen fully through trial and error. Tow cases were evaluated; Case 1 with $Kp = 0.287 Nm/rad$ and $B_p = 0.00065 Nm/rad.s^{-1}$ matched oscillation frequency well but was under damped, and it settle lower than steady state. Case 2 with $Kp = 0.35 Nm/rad$ and $B_p = 0.045 Nm/rad.s^{-1}$ achieved a balanced between frequency , amplitude and decay response with both signals settling at $0.05 rad$ by 65 s.

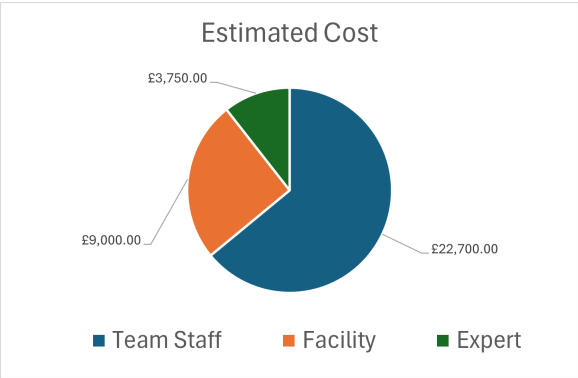
Overall, the model demonstrated a good physical rig response across all three axes.

0.5.2 Cost

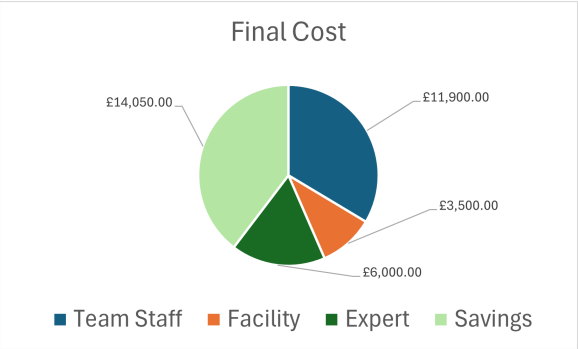
The project cost was generously estimated at the outset to establish a benchmark value that the team aimed to remain within throughout the length of assignment. Staff time was charged at £100/hour, facility usage at £500/hour, and expert consultation at £1000/hour.

As shown in Fig 17, the original estimate projected a total expenditure of £35,450, this full amount is used for part b of the Figure, with the remaining space being presented as Savings. The final recorded total came to £21,400, which resulted in a saving of approximately 40% against the initial estimate (a total of £14,050).

Both the estimate and the final cost of the project fall significantly below of what it would take to finance a physical design of a control system for a twin rotor UAV, which proves the effectiveness in



(a) Estimated Cost - £35,450



(b) Final Cost - £21,400

Figure 17: Comparison between estimated and final project costs.

using tools such as MATLAB & Simulink for design of such systems whilst keeping the overall costs down. This strongly relates to the minimised requirement for physical testing, facility & transport costs, and the need for physical equipment.

0.5.3 Team Analysis

Effective teamwork and time management were a key to maintaining the project's technical quality and ensuring that all deliverables were completed within schedule and budget. This proven to be a challenge due to loosing a team member within week 1, and the second in week 5. It meant that it was vital for the remaining members to carry the workload designed for a team of 5.

It is natural with any workplaces that team members will exhibit different strengths, be able to exceed at different tasks and operate at different levels. This is expected in a well functioning group. However, due to being understaffed there was a significant pressure on the remaining team members to pick up the workload that was intended for a group of 5 people. Resulting from this, what proved far more damaging than being understaffed, is the presence of a contributor whose output cannot be relied upon. When work must be reviewed, corrected, or entirely redone, it consumes time that an already stretched team cannot afford to lose.

In this context an unreliable team member does not represent a neutral absence of contribution, they represent a net negative, actively adding to the workload of those carrying the project. Honesty about capability and availability, even if uncomfortable, is always preferable than the alternative.

Despite these circumstances the two remaining active members adapted continuously, redistributing tasks, covering all technical and reporting deliverables and have maitained the quality of the output throughout the lifetime of the assignment.

Performance Evaluation

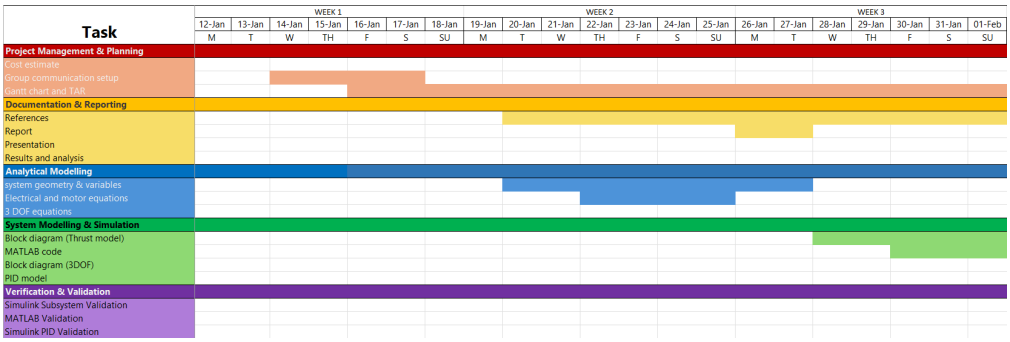
The collaboration between the consistently active team members was strong and sustained throughout the project. Both contributed across all technical and reporting areas, from planning, mathematical modelling, Simulink implementation through to subsystem validation and PID tuning. The third mem-

ber engaged meaningfully during the earlier weeks, contributing to the Block Diagram and Simulink structure, before his involvement diminished significantly from week 8 onwards.

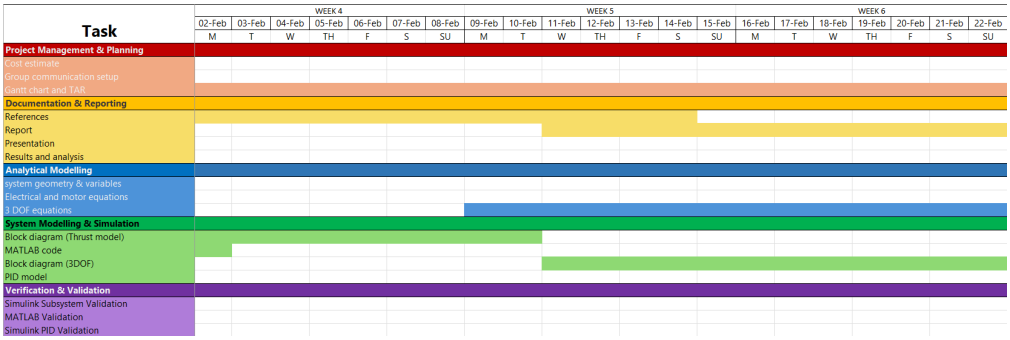
Due to these various circumstances the project ran slightly behind the 10-week schedule, however, the design, implementation, and analysis stages were completed in time for the final presentation and within budget, saving a total of £14,050 as discussed in the cost summary.

Gantt Chart

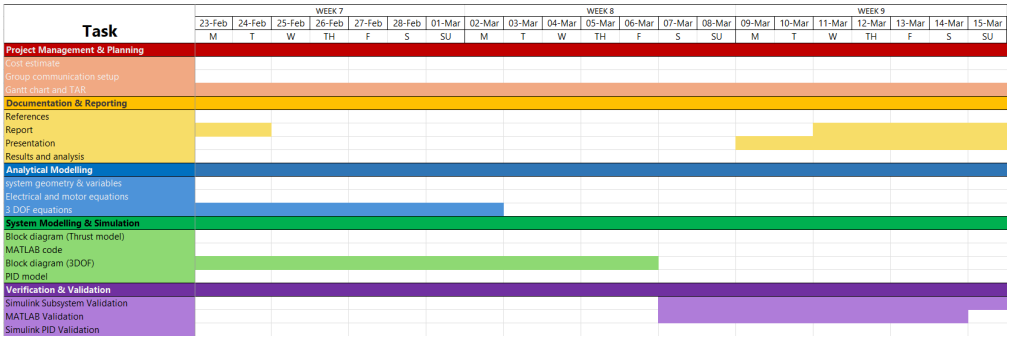
The Gantt chart (Fig 18) reflects the actual progression od the project and validates what was planned at the outset using the V-model (Fig 1). As seen on Fig 18 that the first 5 weeks were predominantly focused on building analytical modelling, mathematical derivation and Simulink implementation, whereas the final weeks shifted towards validation and PID controller design, implementation and testing. The timeline broadly follows the structure anticipated at the start of the assignment, with the final weeks being naturally more compressed as remaining tasks were consolidated by the remaining active team members.



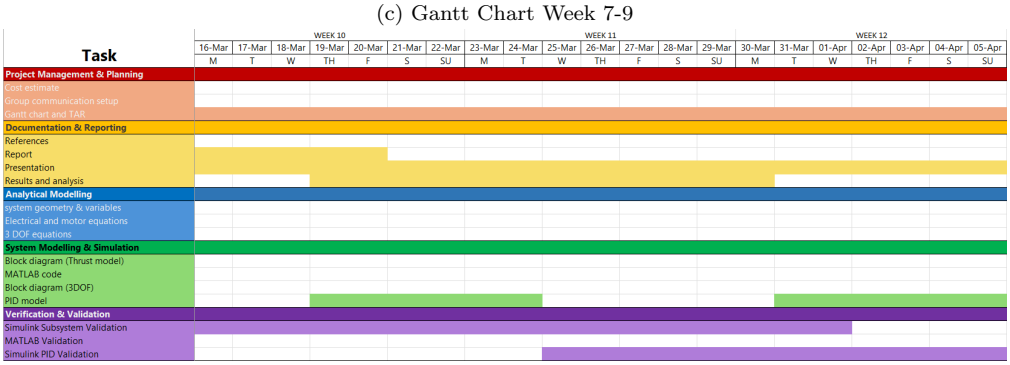
(a) Gantt Chart Week 1-3



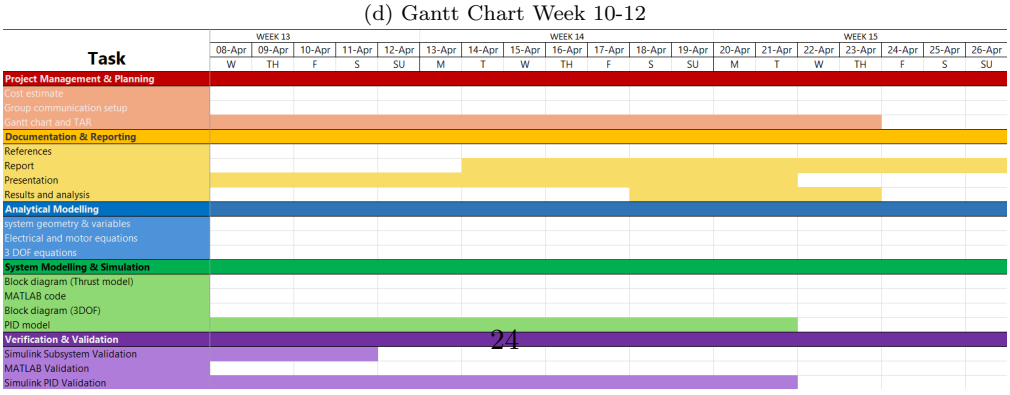
(b) Gantt Chart Week 4-6



(c) Gantt Chart Week 7-9



(d) Gantt Chart Week 10-12



(e) Gantt Chart Week 13-15

Figure 18: Project Gantt chart showing the full workflow.

0.6 Conclusion

The project successfully delivered the specified design requirements of a control system for a twin-rotor UAV rig following the V-model framework from derivation through to controller implementation.

The most significant lesson learnt was the value of establishing the system parameters correctly at the outset. Manufacturer values alone were inadequate, and the time spent correcting this during the validation stage exceeded what a careful identification could have eliminated. Similarly, the Simulink model must be understood and familiarised by the whole team from the start. Each subsystem diagram plays a critical role in the functionality of the whole model, hence understanding how each part translates into the mathematical equation is crucial for troubleshooting and error identification throughout the lifetime of the project.

Simulink proves to be a powerful and indispensable tool for this type of project. Full system validation and PID tuning completed fully in simulation eliminated the need for physical testing, directly reducing development time, facility & component costs, and completely eliminated the risk of equipment failure.